SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05 **Total Marks:100**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I _Shaik Mohammed Abrar / 192472185_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

RUBRICS FOR CALCULATION

TECHNICAL PRESENTATION					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Technical Presentation

Choose ANY ONE Scenario:

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

A. Python programming basics are used
B. Pandas and NumPy are used for data handling
C. Matplotlib and Seaborn are used for result visualization

2. Disease Prediction System

A hospital is planning to build a simple disease prediction system using patient details such as age, blood pressure, sugar level, and cholesterol. Explain how:

A. Python libraries support medical data analysis
B. Scikit-Learn is used for model training and prediction
C. Jupyter Notebook or Google Colab helps in implementation

3. Sales Forecasting Dashboard

A retail company wants to analyze monthly sales data and predict future sales trends. Explain how:

A. Pandas is used for reading and cleaning sales data
B. SciPy and Scikit-Learn are used for analysis and prediction
C. Matplotlib and Seaborn are used to create business visualizations

4. Image Classification Application

A startup is developing an image classification application to identify food items from images. Explain how:

A. Python is used as the development language
B. TensorFlow and PyTorch are used for deep learning model development
C. Notebook environments help in training, testing, and visualizing results

5. Smart Agriculture Data Monitoring

An agriculture research team collects temperature, humidity, soil moisture, and crop growth data from sensors. Explain how:

A. NumPy is used for numerical computation
B. Pandas is used for organizing sensor data
C. Scikit-Learn and visualization libraries support crop condition analysis

TECHNICAL PRESENTATION

1. Student Performance Analysis System
A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:
A. Python programming basics are used
B. Pandas and NumPy are used for data handling
C. Matplotlib and Seaborn are used for result visualization

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling.

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage.

A. Python Programming Basics

Python provides the foundation for developing the Student Performance Analysis System.

1. **Variables** store student details such as name, marks, attendance, and pass percentage.
2. **Data types** like int, float, and string are used to represent student information.
3. **Conditional statements (if-else)** determine whether a student has passed or failed based on marks.
4. **Loops (for, while)** process records of multiple students.
5. **Functions** improve code reusability by calculating average marks, attendance percentage, and pass percentage.
6. **File handling** is used to read and save student data in CSV or text files.

B. Pandas and NumPy for Data Handling

Pandas

Pandas is used to organize and preprocess student data.

1. Load student records from CSV using pd.read_csv().
2. Handle missing values using fillna() or dropna().
3. Remove duplicate records using drop_duplicates().

4. Filter students based on attendance or marks.
5. Calculate average marks, highest marks, lowest marks, and pass percentage.
6. Store data in a **DataFrame** for easy analysis.

NumPy

NumPy performs fast numerical computations.

1. Store marks as arrays.
2. Calculate statistical measures such as:
    - Mean
    - Median
    - Standard Deviation
    - Variance
3. Perform mathematical operations efficiently on large datasets.

C. Matplotlib and Seaborn for Result Visualization

Matplotlib

Matplotlib is used to create basic graphs for student performance.

1. **Bar Chart:** Compare marks of students.
2. **Line Graph:** Show attendance trends over time.
3. **Pie Chart:** Display pass and fail percentages.
4. **Histogram:** Show distribution of student marks.

Seaborn

Seaborn provides advanced statistical visualizations.

1. **Count Plot:** Number of students who passed and failed.
2. **Box Plot:** Detect outliers in student marks.
3. **Heatmap:** Show correlation between marks and attendance.
4. **Scatter Plot:** Analyze the relationship between attendance and marks.

Python Libraries Used

Library	Purpose
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical and statistical calculations
Matplotlib	Basic charts and graphs
Seaborn	Advanced statistical visualizations

Expected Outcome

The Student Performance Analysis System efficiently preprocesses student data, calculates performance statistics, and generates meaningful visualizations. This helps faculty identify high-performing students, students with low attendance, and overall class performance for better academic decision-making.

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


df = pd.read_csv("Student-AT2.csv")


print(df)

print("\nFirst 5 Records")

print(df.head())
```

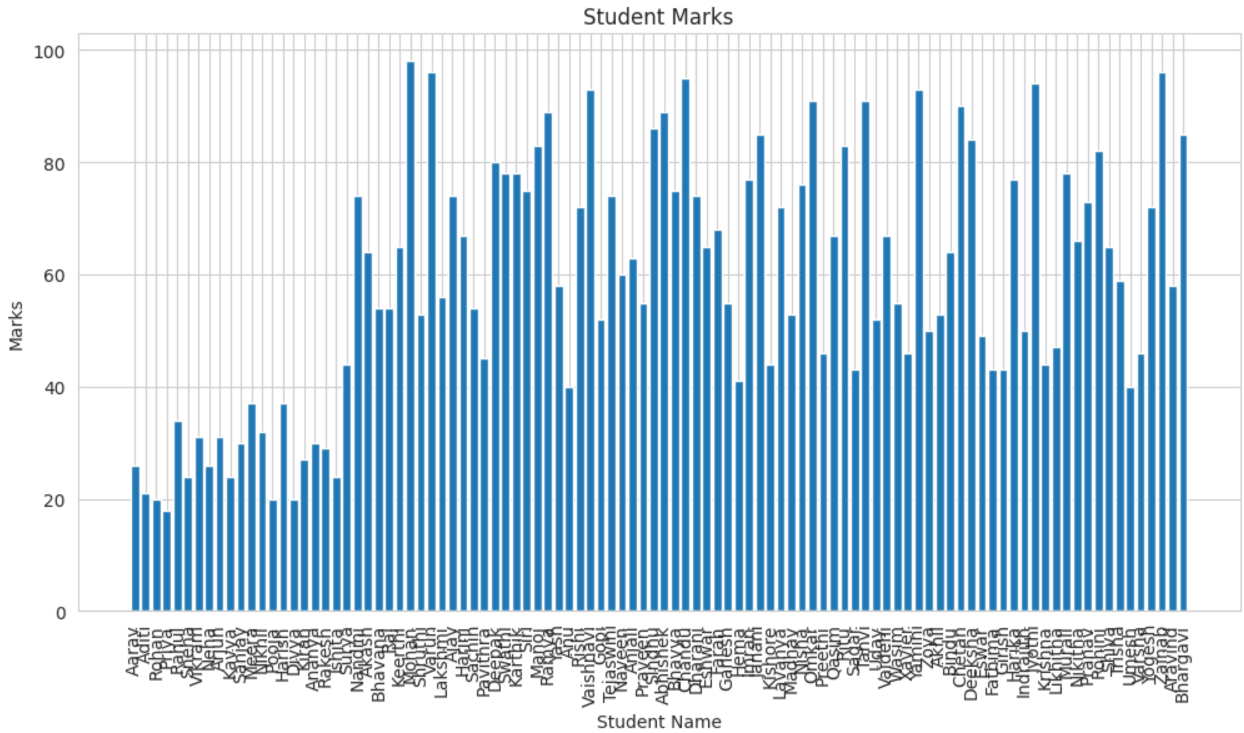
```
print("\nLast 5 Records")
print(df.tail())

print("\nMissing Values")
print(df.isnull().sum())

df = df.drop_duplicates()

cols = ["Attendance", "Assignment_Marks", "Internal_Marks", "Marks"]
df[cols] = df[cols].fillna(df[cols].mean())

df["Result"] = np.where(df["Marks"] >= 40, "Pass", "Fail")

print("\nStatistics")
print("Mean :", df["Marks"].mean())
print("Median :", df["Marks"].median())
print("Mode :", df["Marks"].mode()[0])
print("Std Dev :", df["Marks"].std())
print("Variance :", df["Marks"].var())
print("Maximum :", df["Marks"].max())
print("Minimum :", df["Marks"].min())

print("\nDepartment Wise Average")
print(df.groupby("Department")["Marks"].mean())
```

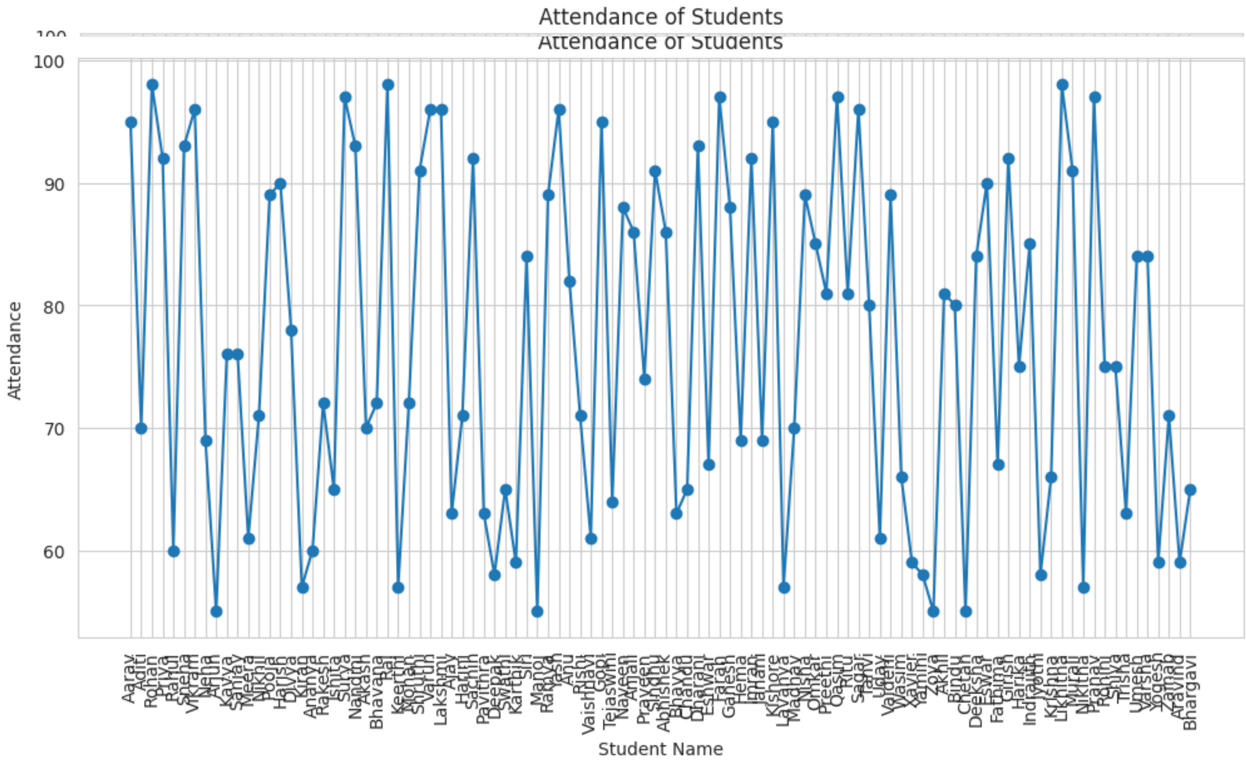
```python
print("\nSemester Wise Average")
print(df.groupby("Semester")["Marks"].mean())

print("\nPass / Fail")
print(df["Result"].value_counts())

print("\nPass Percentage : {:.2f}%".format((df["Result"]=="Pass").mean()*100))
print("Fail Percentage : {:.2f}%".format((df["Result"]=="Fail").mean()*100))

print("\nTop Performer")
print(df.loc[df["Marks"].idxmax()])

print("\nAttendance Below 75")
print(df[df["Attendance"]<75])

sns.set_style("whitegrid")

plt.figure(figsize=(10,5))
plt.bar(df["Name"], df["Marks"])
plt.title("Student Marks")
plt.xticks(rotation=90)
plt.show()

plt.figure(figsize=(10,5))
plt.plot(df["Name"], df["Attendance"], marker="o")
```

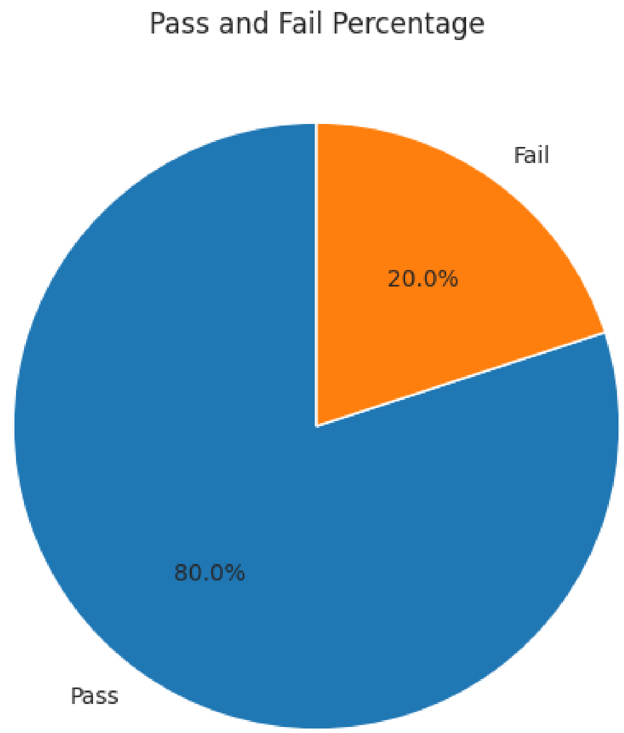
```
plt.title("Attendance")
plt.xticks(rotation=90)
plt.show()

plt.figure(figsize=(5,5))
df["Result"].value_counts().plot.pie(autopct="%1.1f%%")
plt.ylabel("")
plt.show()

plt.figure(figsize=(6,4))
plt.hist(df["Marks"], bins=10)
plt.title("Marks Distribution")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Attendance"], df["Marks"])
plt.title("Attendance vs Marks")
plt.show()

sns.boxplot(y=df["Marks"])
plt.show()

sns.countplot(x="Department", data=df)
plt.show()
```

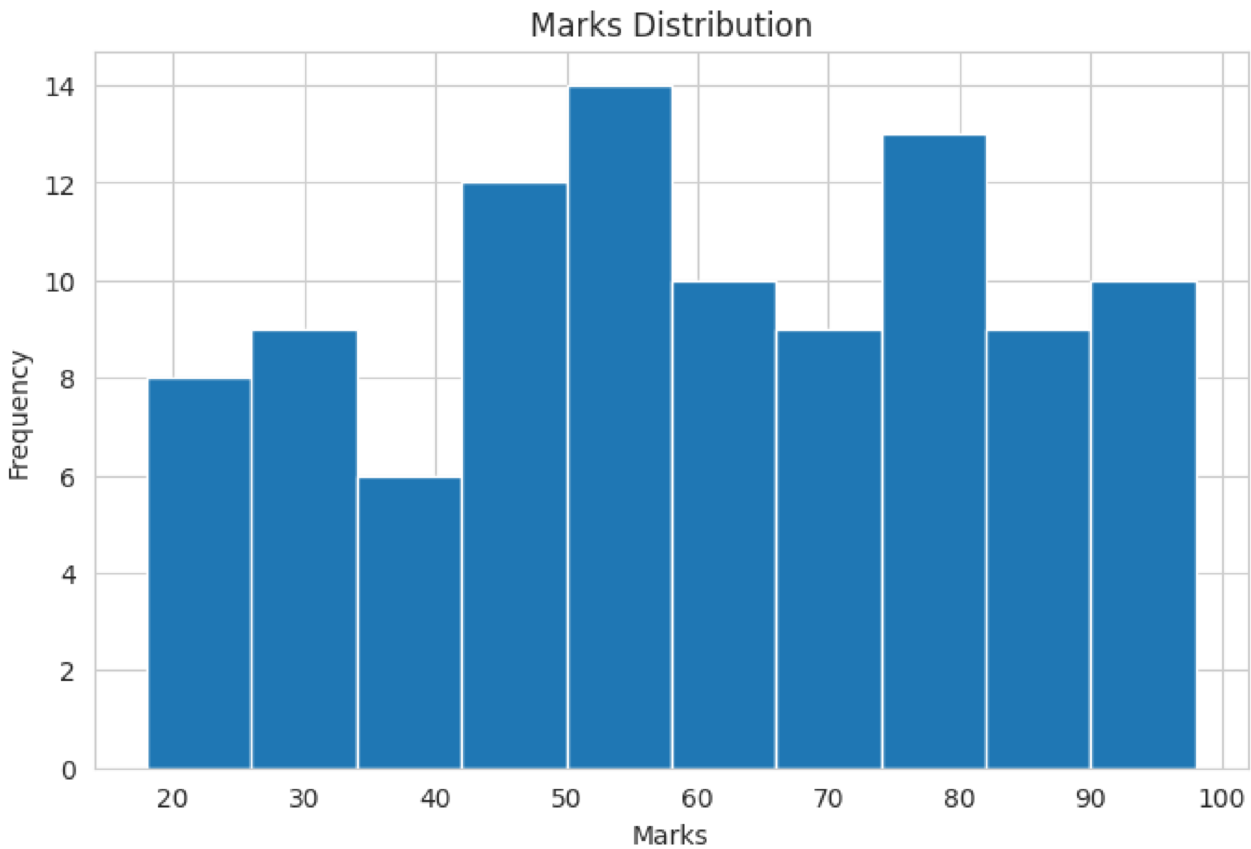
```
sns.countplot(x="Result", data=df)

plt.show()


sns.barplot(x="Department", y="Marks", data=df)

plt.show()


sns.boxplot(x="Department", y="Marks", data=df)

plt.show()


sns.scatterplot(x="Attendance", y="Marks", hue="Department", data=df)

plt.show()


sns.heatmap(df[cols].corr(), annot=True, cmap="coolwarm")

plt.show()


print("\nAnalysis Completed Successfully.")
```

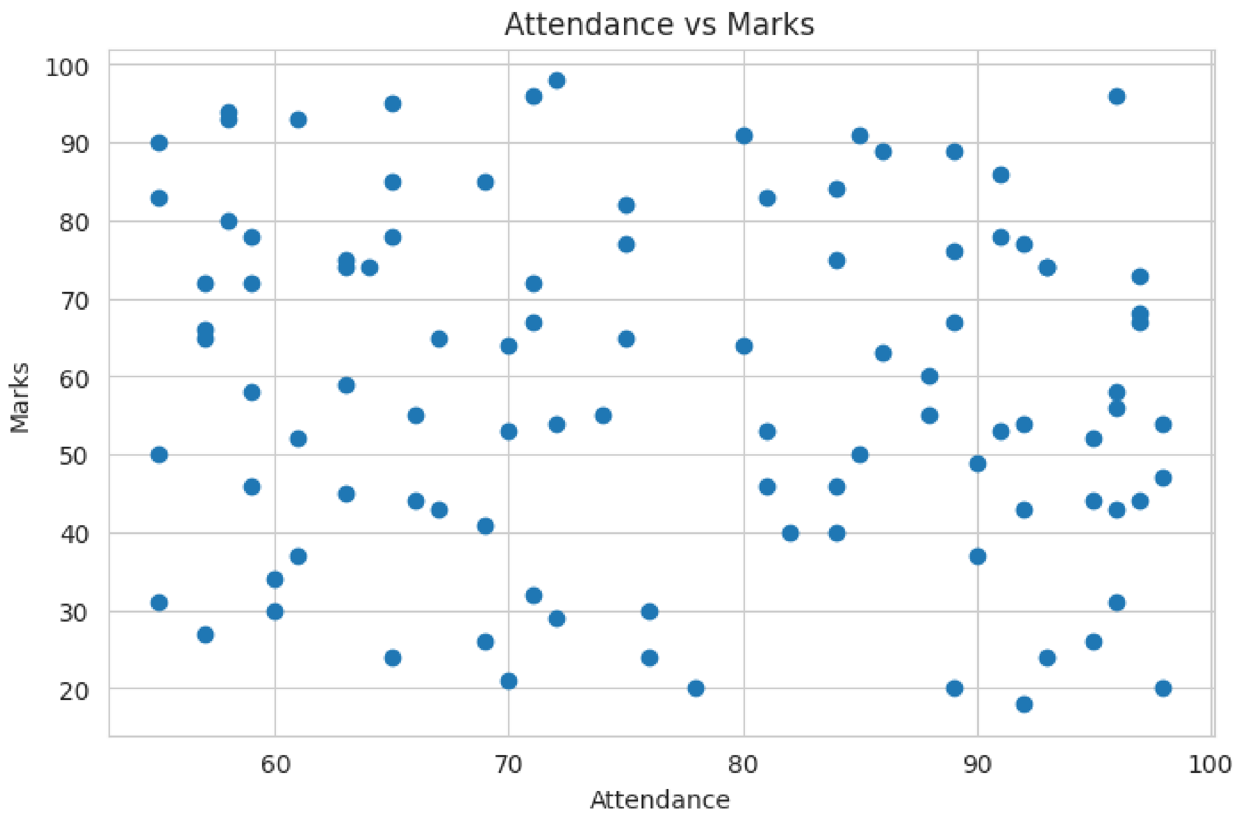
Output 1:

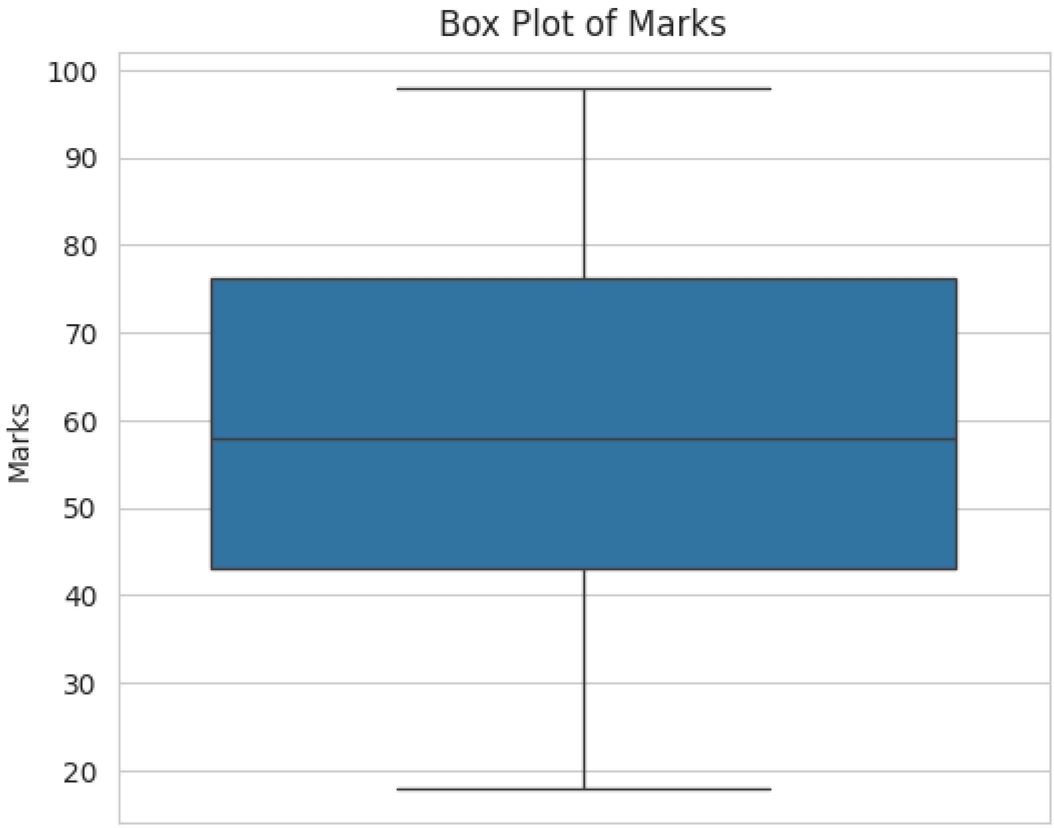

Student Marks
Marks
0
20
40
60
80
100
Student Name

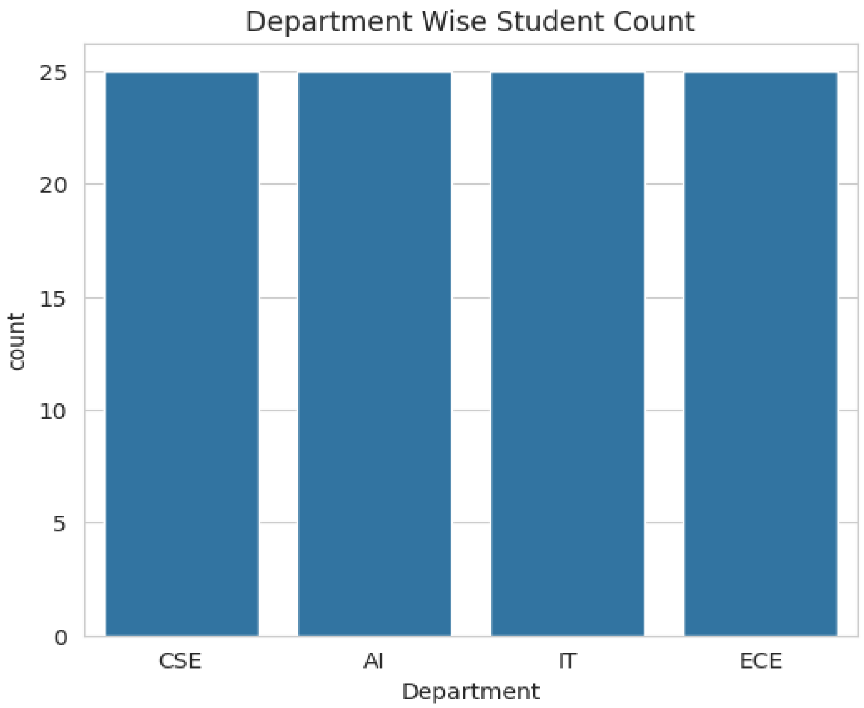

Attendance of Students
Attendance of Students
Attendance
60
70
80
90
100
Student Name

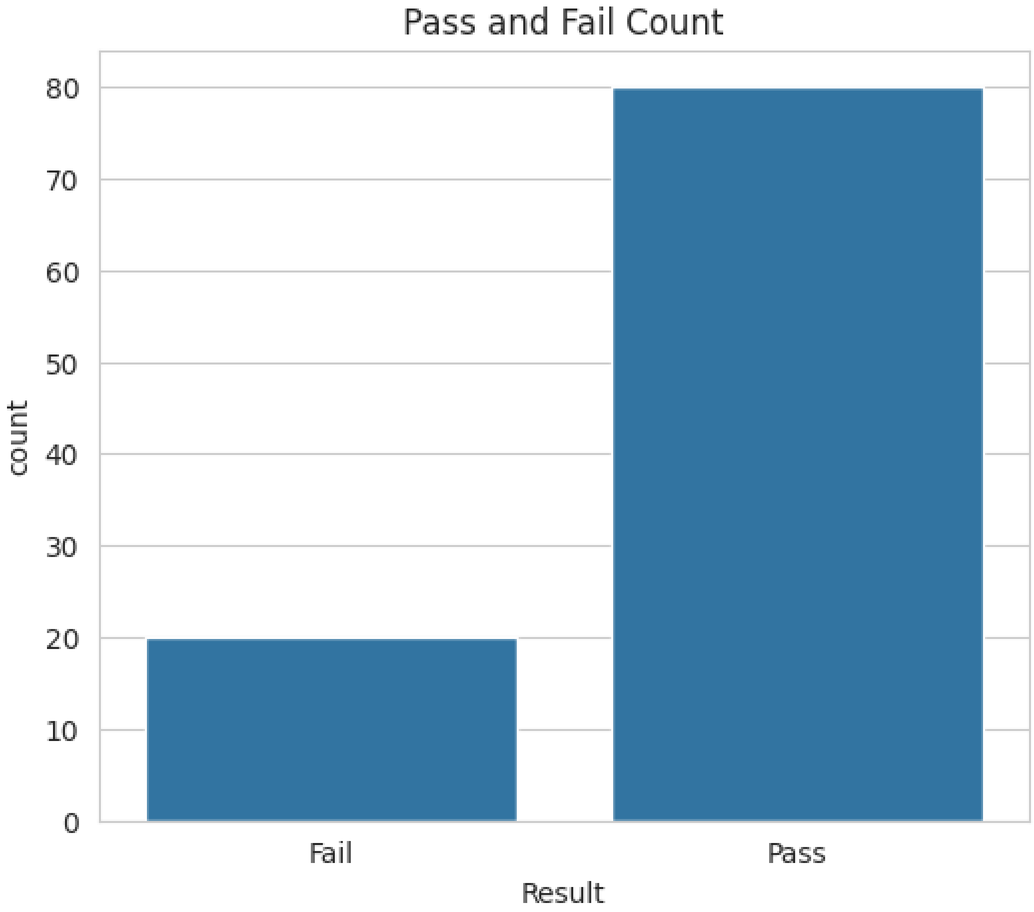

Pass and Fail Percentage
Fail
20.0%
80.0%
Pass

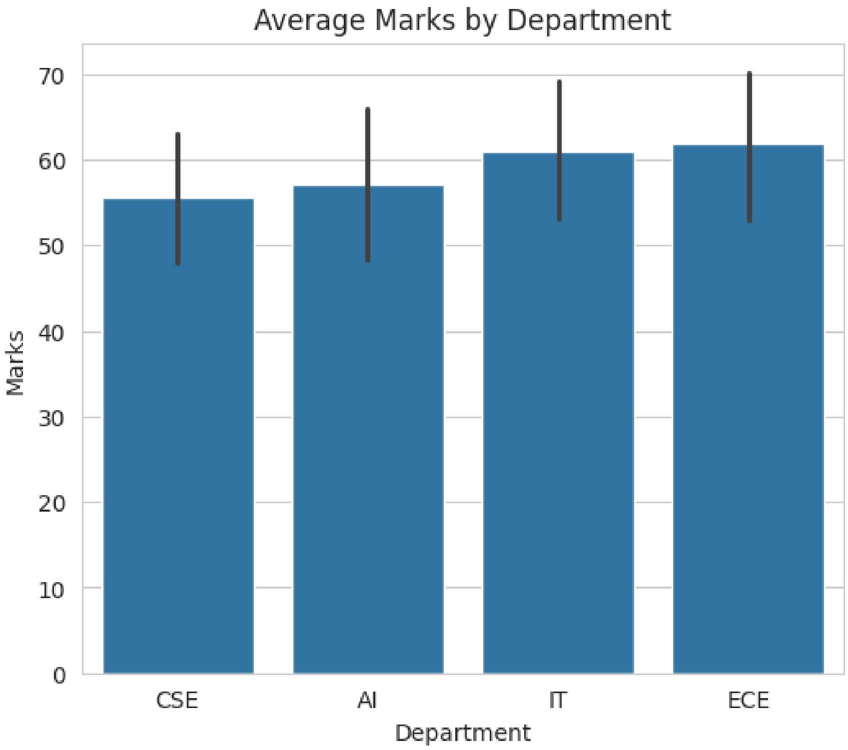

Marks Distribution
Frequency
0
2
4
6
8
10
12
14
20
30
40
50
60
70
80
90
100
Marks


Attendance vs Marks
Marks
Attendance
100
90
80
70
60
50
40
30
20
60
70
80
90
100

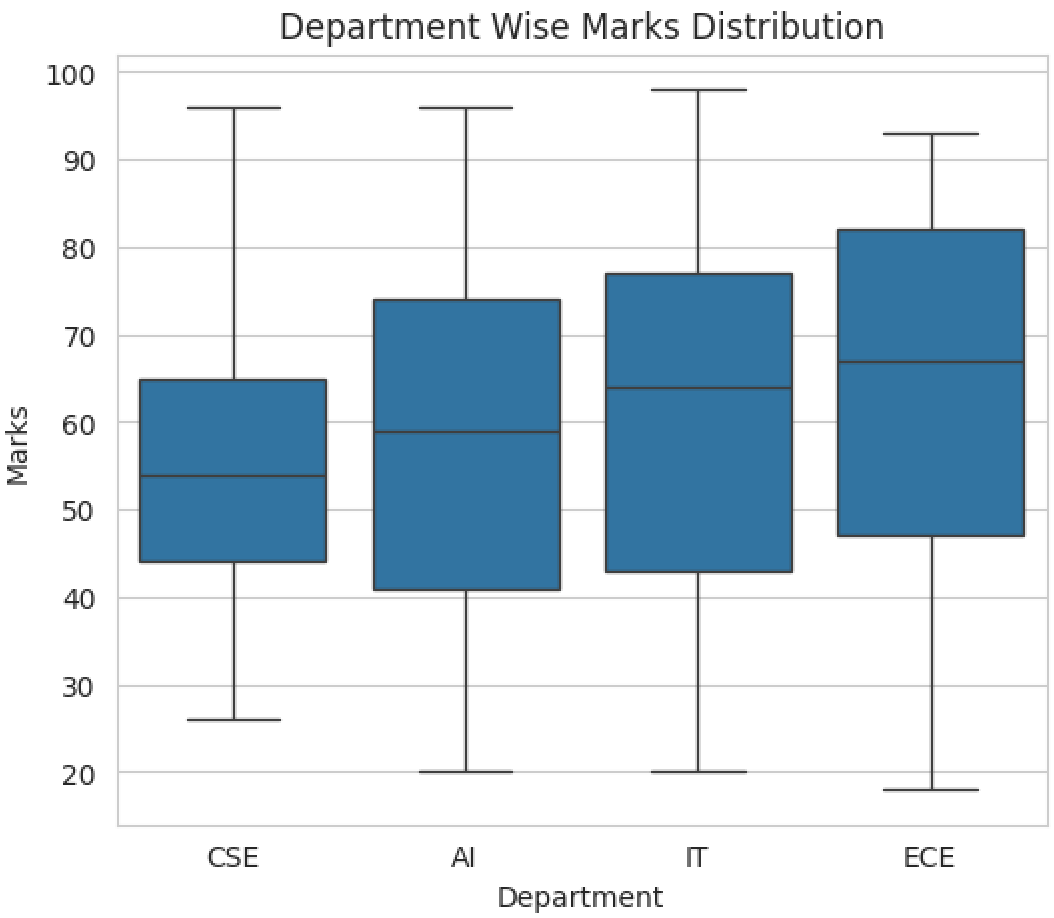

Box Plot of Marks
Marks
100
90
80
70
60
50
40
30
20


Department Wise Student Count
count
25
20
15
10
5
0
CSE
AI
IT
ECE
Department

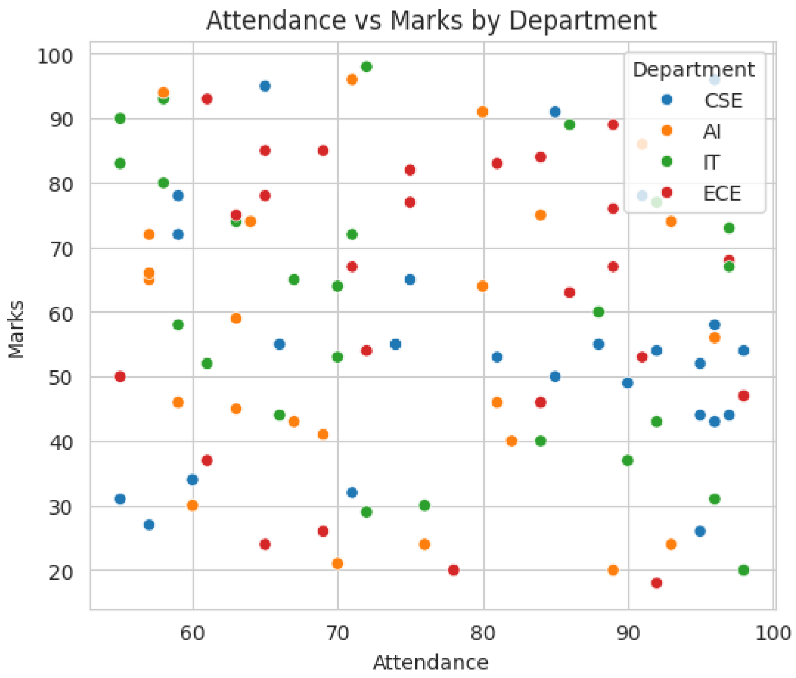

Pass and Fail Count
count
80
70
60
50
40
30
20
10
0
Fail
Pass
Result



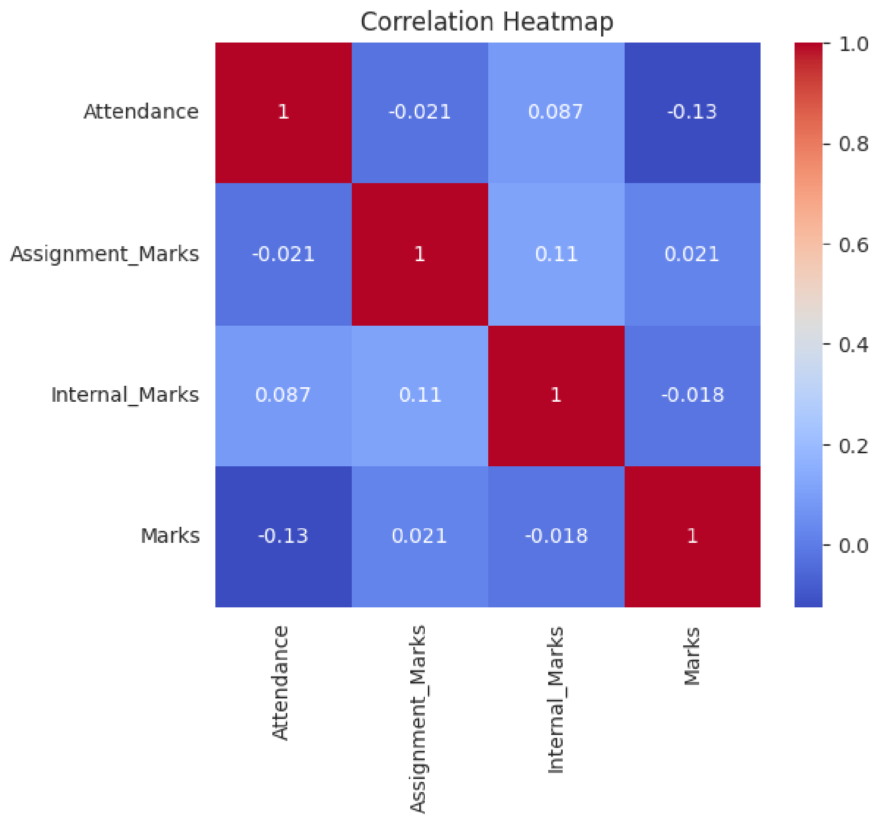
Average Marks by Department
Marks
70
60
50
40
30
20
10
0
CSE
AI
IT
ECE
Department



Department Wise Marks Distribution
Marks
100
90
80
70
60
50
40
30
20
CSE
AI
IT
ECE
Department



Attendance vs Marks by Department
Department
CSE
AI
IT
ECE
Marks
Attendance
100
90
80
70
60
50
40
30
20
60
70
80
90
100



Correlation Heatmap
Attendance
Assignment_Marks
Internal_Marks
Marks
1
-0.021
0.087
-0.13
-0.021
1
0.11
0.021
0.087
0.11
1
-0.018
-0.13
0.021
-0.018
1
1.0
0.8
0.6
0.4
0.2
0.0